

SOURAV KUMAR

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EDUCATION

University of Petroleum and Energy Studies (UPES), Dehradun

B. Tech, Computer Science & Engineering

May 2026

CGPA: 7.36/10

SKILLS

Programming Languages : Java, Kotlin, C, C++, HTML/CSS, Java Script, Python, Matlab,R,GO,SQL

Framework & Tools : Scikit-learn, Supervised Learning (Regression, Classification), Linear Regression, Decision Tree, OpenCV, Langchain, LLM, NLP, Hugging Face, RAG, Transformer, YOLO, Resnet,Fiinetuning, Pandas, NumPy, Matplotlib,Seaborn, Exploratory Data Analysis (EDA) AWS (S3, EC2, Lambda), Jekins, Docker, Google Colab, Jupyter Notebook, Flask, Django, React.js, Node.js, Android Studio, Jupyter Notebook, Google Colab, Git, GitHub, VS Code

Model Evaluation: MAE, MSE, RMSE, R² Score, Feature Engineering, Hyperparameter Tuning

EXPERIENCE

Intern | IBM | Remote

June 2025 - July 2025

- Assisted in collecting and preprocessing structured healthcare datasets. Supported exploratory data analysis (EDA) to uncover cost-driving factors.
- Aided in implementing regression models including linear regression, decision tree, and random forest.
- Built a predictive system to estimate medical insurance charges.
- Visualized insights using Matplotlib/Seaborn.

Freelanc | Fivver & Upwork | Remote

June 2025 - Present.

- Designed, developed, and deployed high-performance responsive websites for educational institutions, retail businesses, and SMEs across diverse industries.
- Delivered customized web solutions that enhanced digital presence, optimized user experience, and increased customer engagement and brand visibility.
- Collaborated with international clients across diverse regions to gather requirements, deliver customized web solutions, and manage end-to-end project execution.

Social Work Intern | Samadhaan Foundation

June 2024 - Aug 2024

- Assisted in community outreach programs.
- Conducted data analysis and helped in digital management initiatives.

PROJECTS

Stock Prediction Models

Aug 2025.

- Developed multiple stock price prediction models using Machine Learning and Deep Learning techniques, improving forecasting accuracy by evaluating and comparing model performance across algorithms such as LSTM, ARIMA, and Prophet.
- Performed data preprocessing, feature engineering, and normalization to improve prediction accuracy.
- Implemented visualization dashboards using Matplotlib and Seaborn for exploratory data analysis (EDA), enabling insights into historical trends, volatility, and model performance metrics.
- Conducted hyperparameter tuning and cross-validation to optimize models, resulting in more stable forecasting outcomes and reduced overfitting across test scenarios.

Enterprise-FraudShield-AI

Dec 2024

- Engineered a real-time enterprise fraud detection system using Python, FastAPI, and Machine Learning models to identify suspicious transactions, refund abuse, and anomalous customer activities with scalable risk-scoring pipelines.
- Integrated Apache Kafka for high-throughput transaction streaming and Redis for ultra-fast caching, enabling low-latency fraud monitoring and improving real-time processing performance across distributed services.
- Developed a production-ready backend architecture with FastAPI, Docker, and Uvicorn, implementing secure REST APIs, asynchronous processing, and microservices-ready deployment for scalable enterprise environments.
- Implemented fraud prediction and behavioral analytics using Scikit-learn, XGBoost, and anomaly detection techniques, optimizing transaction pattern analysis and enhancing fraud detection accuracy for large-scale financial systems.

NeuroFin - Classification Tool (Transfer Learning)

May 2024

- Trained an image classification model using transfer learning with MobileNetV3-Large and a dataset of 9,000 images to accurately classify 9 fish species with 90% overall accuracy and reducing reliance on manual inspection.
- Enhanced model robustness and usability through Python-based preprocessing, augmentation pipelines, and splitting (80/20 train-validation) ensuring scalable deployment across varied datasets.

CERTIFICATIONS

• Data Structures and Algorithms (DSA) Certification – Coursera	2025
• IBM Certified Internship – Healthcare Cost Prediction	2025
• Python Programming – CodeWithHarry (Self Learning)	2023
• Harvard CS50AI: CS50’s Introduction to Artificial Intelligence with Python	2024
• Microsoft Certified: Azure AI Fundamentals (AI-900)	2024

KEY ACHIEVEMENTS

• Member of the Coding Club at UPES	2022 -2026
• Participated in elite coding contests on LeetCode and CodeChef, consistently strengthening algorithmic problem-solving under timed conditions.	2025
• Contributed in university and national hackathons by leading innovative AI/product strategies, rapid prototyping, and high impact team execution.	2025
• Participated twice in CSI Hackathons, contributing innovative solutions with a team-based approach and got prize	2024
• Participated in Smart India Hackathon 2024, building scalable real-world solutions with a high-performing collaborative team.	2024
• Volunteered at TechFest 2023, UPES	2023
• Active participant in hackathons and coding challenges	2023,2024,2025